Minor Project

Understanding DNA Damage and Repair

Topics Connected

- Genes and Proteins
- DNA Repair & Gene Silencing

Aim

To understand how changes or damage to DNA can affect genes and proteins, and how cells use DNA repair mechanisms to maintain genome stability.

Student Task

Choose **one well-documented genetic disorder or mutation example** and explain:

1. **What is the gene?**
2. **What does the normal gene/protein do?**
3. **What type of DNA mutation is involved?**
4. **How can the mutation affect the protein?**
5. **What biological/phenotypic effect can result?**
6. **Why is DNA repair important?**
7. Identify the **relevant DNA repair mechanism(s)** at a basic level.
8. Explain the difference between **HR and NHEJ**.

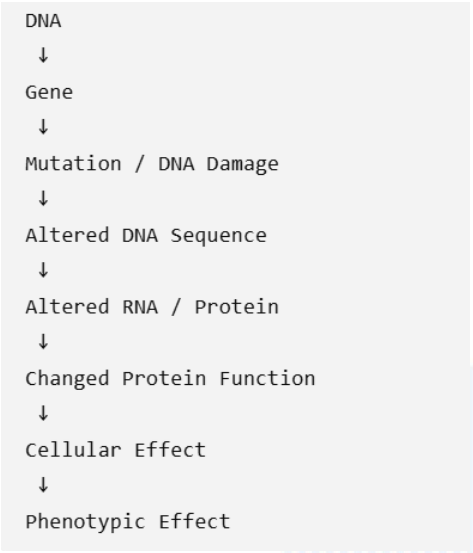
Suggested Examples

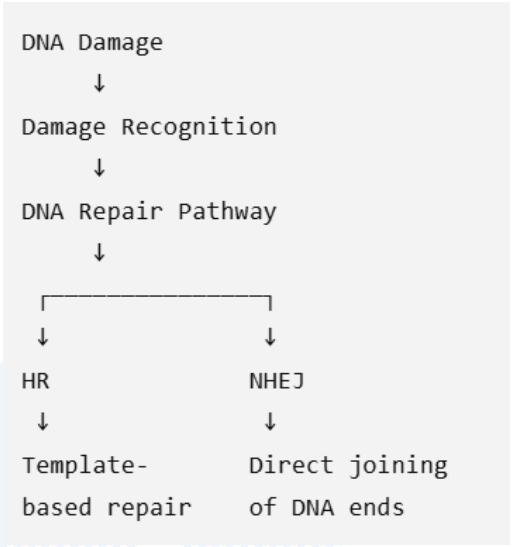
Students can choose one:

- **Sickle cell disease — HBB**
- **Cystic fibrosis — CFTR**
- **Xeroderma pigmentosum — DNA repair genes**
- **Lynch syndrome — mismatch repair genes**
- **BRCA1/BRCA2-associated cancer predisposition**

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Required Flowchart





Required Sections

Section 1 — Introduction

Briefly introduce the selected genetic disorder/mutation.

Section 2 — Normal Gene and Protein

Explain:

- Gene name
- Protein produced
- Normal function

Section 3 — Mutation

Explain:

- What is a mutation?
- What type of mutation occurs in your example?
- Where does it occur?

A detailed molecular mechanism is **not required**.

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Section 4 — Effect on Protein

Explain:

DNA change → RNA/protein change → protein-function change

Students can use a simple diagram.

Section 5 — DNA Repair

Briefly explain:

- Why DNA gets damaged
- Why repair is necessary
- BER
- NER
- MMR
- HR
- NHEJ

Students **do not need to explain the detailed molecular steps** of every pathway.

Section 6 — HR vs NHEJ

Section 7 — Conclusion

In **one short paragraph**, explain the connection:

> **DNA integrity → gene sequence → protein → cellular function → phenotype**

and why DNA repair is important for maintaining genome stability.

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8. Submission Requirements

- **Length:** 3–4 pages
- **Font:** Times New Roman
- **Body:** 12 pt
- **Headings:** 14–16 pt, bold
- **At least:** 1 flowchart + 1 comparison table
- **References:** 3–4 reliable scientific sources
- **Citations:** Cite information below the relevant diagram/table or in the text.
- **Images:** Any external image must have a caption and citation.
- **No advanced experimental procedures required.**

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